

Notes for ‘The unit circle and trigonometry’

Important Ideas and Useful Facts:

- (i) **Angles, degrees and radians:** Angles in the plane are *positive* when measured *anticlockwise*, and *negative* when measured *clockwise*. Angles may be measured in *degrees* (using the symbol $^\circ$) or in *radians* (expressed usually simply as real numbers).

The angle of a full circle is 360° . The *radian* measure of an angle θ is the length of an arc on a unit circle subtended by θ . In particular 360° corresponds to 2π radians, the perimeter of the unit circle. Thus 1° corresponds to $\frac{2\pi}{360} = \frac{\pi}{180}$ radians, and one radian corresponds to $\frac{180}{\pi}$ degrees, which is approximately 57.3° .

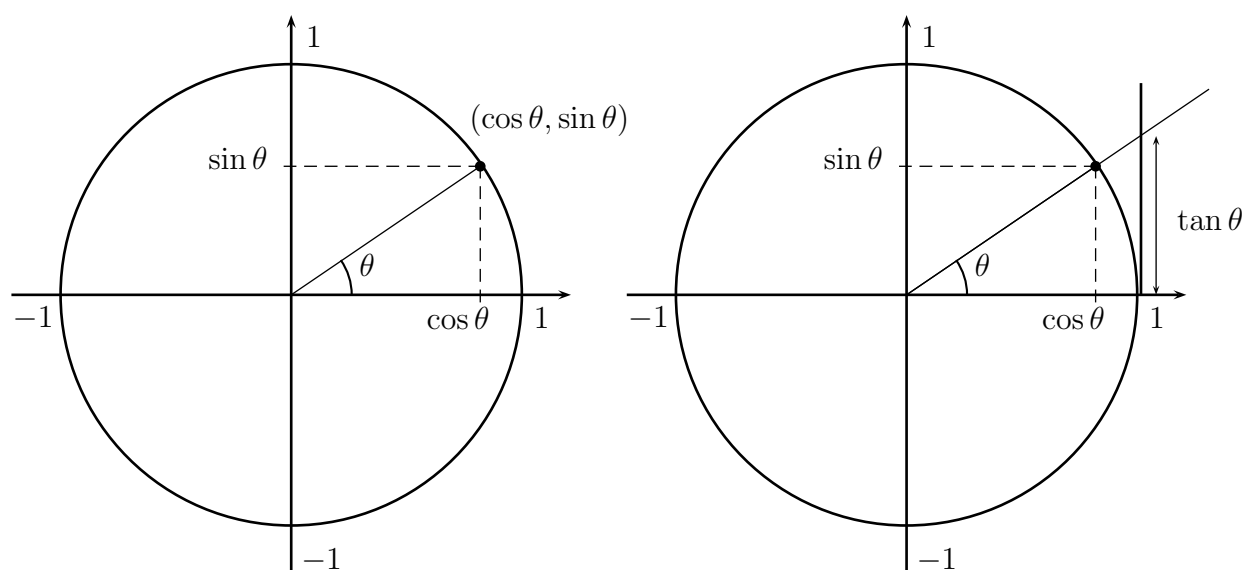
Some common conversions between degrees and radians are as follows:

degrees	1°	30°	45°	60°	90°	120°	135°	150°	180°	270°	360°
radians	$\frac{\pi}{180}$	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{3\pi}{4}$	$\frac{5\pi}{6}$	π	$\frac{3\pi}{2}$	2π

- (ii) **The sine, cosine and tangent of any angle:** For an arbitrary angle θ , we put

$$(\cos \theta, \sin \theta) = (x, y) \quad \text{and} \quad \tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{y}{x} \quad (\text{provided } x = \cos \theta \neq 0),$$

where (x, y) lies on the unit circle satisfying the equation $x^2 + y^2 = 1$, centred at the origin $(0, 0)$, and where the radius from $(0, 0)$ to (x, y) subtends an angle θ :



The point on the circle can be in any quadrant. Note, however, that for an acute angle θ , where the point on the unit circle ends up in the first quadrant, as in the diagram above, these definitions coincide with definitions given previously. To see this, one uses ratios of sides of the right-angled triangle formed using the point on the circle, the origin and the point on the horizontal axis labelled by $\cos \theta$. The radius of length 1 becomes the hypotenuse.

It follows also, by considering similar right-angled triangles, that $\tan \theta$ is the length of the line segment along the tangent to the circle, from the x -axis to the intersection point with the line extending the radius, measured negatively in the 2nd and 4th quadrants.

(iii) **Equivalent angles:** Angles are *equivalent* if they differ by integer multiples of 2π . For example, $0, \pm 2\pi, \pm 4\pi, \dots$ are equivalent; $\frac{\pi}{6}, -\frac{11\pi}{6}$ and $\frac{13\pi}{6}$ are equivalent; $-\frac{\pi}{2}, \frac{3\pi}{2}$ and $\frac{7\pi}{2}$ are equivalent. The values of $\sin$, $\cos$ and $\tan$ remain the same for equivalent angles.

(iv) **Common trig values for first quadrant angles:**

$$\begin{aligned}\sin 0 &= 0, & \cos 0 &= 1, & \tan 0 &= 0, & \sin \frac{\pi}{6} &= \frac{1}{2}, & \cos \frac{\pi}{6} &= \frac{\sqrt{3}}{2}, & \tan \frac{\pi}{6} &= \frac{1}{\sqrt{3}}, \\ \sin \frac{\pi}{4} &= \frac{1}{\sqrt{2}}, & \cos \frac{\pi}{4} &= \frac{1}{\sqrt{2}}, & \tan \frac{\pi}{4} &= 1, & \sin \frac{\pi}{3} &= \frac{\sqrt{3}}{2}, & \cos \frac{\pi}{3} &= \frac{1}{2}, & \tan \frac{\pi}{3} &= \sqrt{3}, \\ \sin \frac{\pi}{2} &= 1, & \cos \frac{\pi}{2} &= 0, & \tan \frac{\pi}{2} & \text{ is undefined.}\end{aligned}$$

(v) **Signs of trig values inside the respective quadrants:** In the first quadrant (not including the axes), we have

$$\sin \theta > 0, \quad \cos \theta > 0, \quad \tan \theta = \frac{\sin \theta}{\cos \theta} > 0.$$

In the second quadrant (not including the axes), we have

$$\sin \theta > 0, \quad \cos \theta < 0, \quad \tan \theta = \frac{\sin \theta}{\cos \theta} < 0.$$

In the third quadrant (not including the axes), we have

$$\sin \theta < 0, \quad \cos \theta < 0, \quad \tan \theta = \frac{\sin \theta}{\cos \theta} > 0.$$

In the fourth quadrant (not including the axes), we have

$$\sin \theta < 0, \quad \cos \theta > 0, \quad \tan \theta = \frac{\sin \theta}{\cos \theta} < 0.$$

(vi) **Trig values on the axes:** Points on the unit circle with angles θ equivalent to $0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$ fall on the axes at the points $(1, 0), (0, 1), (-1, 0), (0, -1)$ respectively. We therefore get the following trig values:

$$\begin{aligned}\sin 0 &= 0, & \cos 0 &= 1, & \tan 0 &= 0, \\ \sin \left(\frac{\pi}{2}\right) &= 1, & \cos \left(\frac{\pi}{2}\right) &= 0, & \tan \left(\frac{\pi}{2}\right) & \text{ is undefined,} \\ \sin \pi &= 0, & \cos \pi &= -1, & \tan \pi &= 0, \\ \sin \left(\frac{3\pi}{2}\right) &= -1, & \cos \left(\frac{3\pi}{2}\right) &= 0, & \tan \left(\frac{3\pi}{2}\right) & \text{ is undefined.}\end{aligned}$$

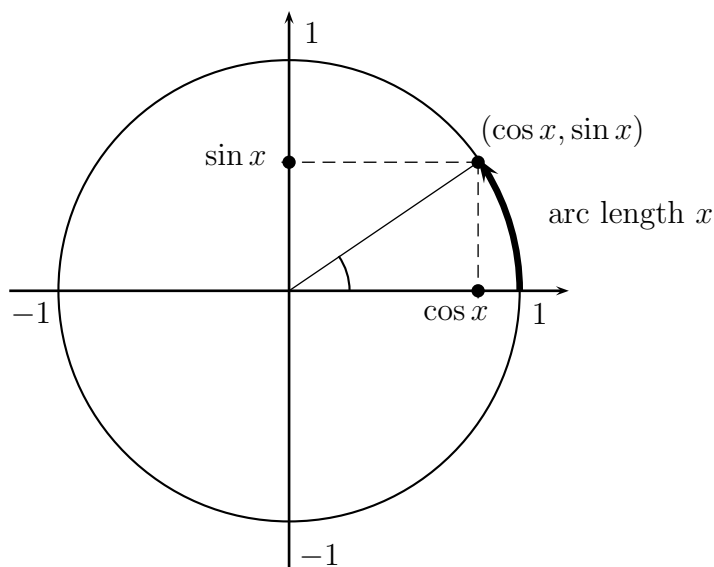
(vii) **The circular functions:** Finally we can define the *circular functions*. However, we need to change our point of view slightly, in order to use x as an input.

In the preceding discussion we have used θ to denote an angle associated with the unit circle in the xy -plane.

To use x as an input, we ignore the labels on the axes, and think of x now as the arc length subtended by an angle on the point of the unit circle, centred at the origin, and the angle as usual is measured in the positive direction by moving anticlockwise from the point 1 on the horizontal axis.

Thus $\sin x$ is the real number obtained by moving directly across to the vertical axis, and $\cos x$ is the real number obtained by moving directly up or down to the horizontal axis.

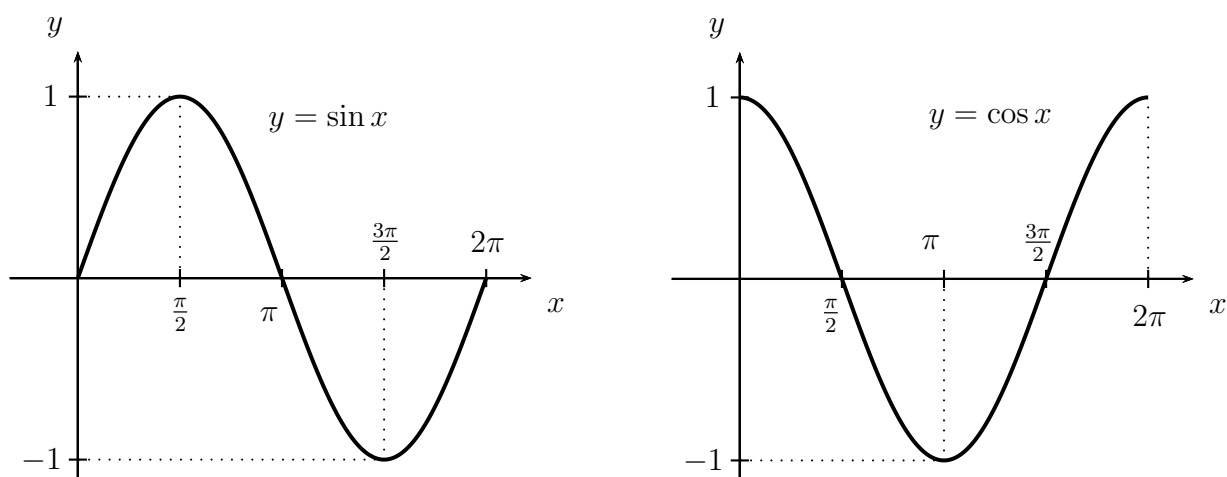
(Exactly how one moves across to the right or left to the vertical axis, or up or down to the horizontal axis, depends on the quadrant of the circle in which the point happens to be. In the following diagram the point $(\cos x, \sin x)$ happens to be in the first quadrant.)



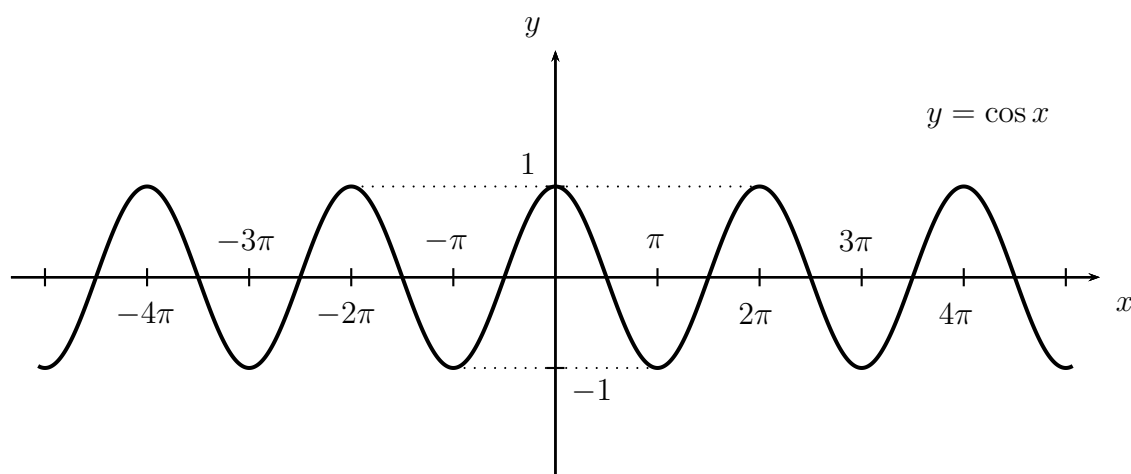
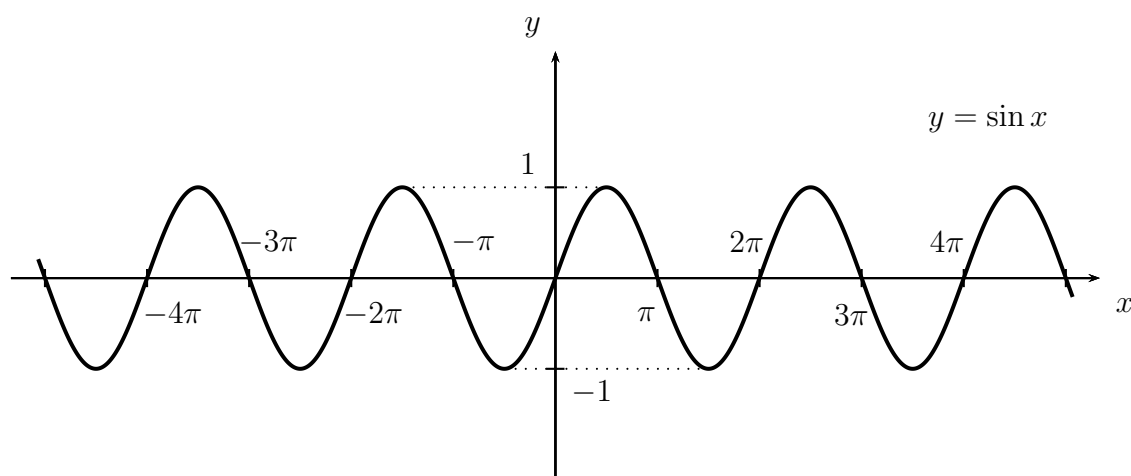
Consider moving anticlockwise one revolution of the circle, but broken up into four stages:

- (1) as x goes from 0 to $\frac{\pi}{2}$ in the first quadrant, $\sin x$, the point on the vertical axis, goes from 0 to 1, whilst $\cos x$, the point on the horizontal axis, goes from 1 to 0.
- (2) as x goes from $\frac{\pi}{2}$ to π in the second quadrant, $\sin x$ returns from 1 to 0, whilst $\cos x$ goes from 0 to -1 .
- (3) as x goes from π to $\frac{3\pi}{2}$ in the third quadrant, $\sin x$ goes from 0 to -1 , whilst $\cos x$ returns from -1 to 0.
- (4) as x goes from $\frac{3\pi}{2}$ to 2π in the fourth quadrant, $\sin x$ returns from -1 to 0, whilst $\cos x$ goes from 0 to 1.

We can plot the behaviour of $y = \sin x$ and $y = \cos x$ for this single anticlockwise revolution of the circle, now in the xy -plane, for $0 \leq x \leq 2\pi$:



We now allow the angle x , measured as arc length, to make unlimited numbers of revolutions of the unit circle both clockwise and anticlockwise and we produce curves that oscillate infinitely often, in both the positive and negative directions along the entire real number line:



The functions $y = \sin x$ and $y = \cos x$ we produce in this way are called the *circular functions*, because the pair $(\cos x, \sin x)$ form coordinates on the unit circle. They both have the same domain, which is the entire real line $\mathbb{R}$, and the same range, which is the interval $[-1, 1]$.

Their graphs are called *sinusoidal curves*. Notice that the sine and cosine curve have exactly the same shape. One can get the cosine curve from the sine curve by translating the sine curve horizontally to the left by $\frac{\pi}{2}$ units, so that

$$\cos x = \sin \left(x + \frac{\pi}{2} \right),$$

and we can get the sine curve from the cosine curve by translating the cosine curve horizontally to the right by $\frac{\pi}{2}$ units, so that

$$\sin x = \cos \left(x - \frac{\pi}{2} \right).$$